

CAPSTONE PROJECT 2

Chapter 5 Additional Writing Guidelines

CHAPTER 5 includes the SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS.

- A simple way to remember the three sections is:

Summary = What happened?

Conclusions = What do the results mean?

Recommendations = What should be done next?

5.1 SUMMARY

A. What is the Summary?

- The Summary provides a brief but complete overview of the entire capstone project.
- It should summarize the important information presented in Chapters 1-4 without repeating the entire contents of the previous chapters.

The summary should normally include:

1. The problem addressed
2. The purpose of the project
3. The intended users or beneficiary
4. The objectives
5. The development methodology
6. The major features of the system
7. The implementation
8. The testing performed
9. The major evaluation results

B. What Should NOT Be Included?

- Avoid putting new information in the Summary.
- **For Example**, if Chapter 4 did not discuss mobile compatibility, students should not suddenly say in Chapter 5:

"The system was also successfully tested on Android devices."

If that testing was not presented in Chapter 4, it should not suddenly appear in Chapter 5.

Rule: Chapter 5 summarizes what was already established in Chapters 1-4.

C. Recommended Structure of the Summary

A useful structure is:

Paragraph 1: Problem and Purpose

- Explain what problem the project addressed.

Paragraph 2 : Objectives and Development

- Explain what the project intended to accomplish and how the system was developed.

Paragraph 3 : Major System Features

- Briefly describe the completed system.

Paragraph 4 : Testing and Evaluation

- Discuss how the system was tested and evaluated.

Paragraph 5 : Overall Result

- State whether the project achieved its intended purpose.

Example Scenario:

Suppose the students developed:

"Web-Based Inventory and Sales Management System for ABC Hardware Store"

Problem :

The hardware store was using:

- handwritten inventory records;
- Excel spreadsheets;
- manual sales computation; and
- manually prepared reports.

This caused:

- inaccurate inventory records;
- slow report generation;
- difficulty monitoring stocks;
- duplicate records; and
- computation errors.

Proposed Solution

The students developed a web-based system containing:

- Login

- Dashboard
- Product Management
- Inventory Management
- Sales
- Supplier Management
- Reports
- User Management

Sample 5.1 Summary

The project focused on the development of a Web-Based Inventory and Sales Management System for ABC Hardware Store to address the problems associated with its existing manual inventory and sales processes. The existing process required employees to manually record product transactions, monitor stock levels, compute sales, and prepare reports, which resulted in delays, recording errors, and difficulties in monitoring inventory.

The primary objective of the project was to develop a computerized system that would automate inventory management, sales recording, stock monitoring, and report generation. The system was designed for the store administrator, manager, and authorized employees. The researchers gathered the requirements of the organization and used these requirements as the basis for designing and developing the proposed system.

The system was developed using PHP, Laravel, MySQL, HTML, CSS, and JavaScript. The major modules included user authentication, dashboard, product management, inventory management, sales management, supplier management, and report generation. These modules were implemented to provide an integrated solution for the organization's daily operations.

After implementation, the system underwent functional testing to determine whether its major features performed according to the specified requirements. User evaluation was also conducted to determine the system's acceptability and software quality based on the selected evaluation criteria. The testing results showed that the major functions of the system operated as expected, while the evaluation results indicated that the system was acceptable to its intended users.

Overall, the project successfully produced a functional computing solution that addressed the identified problems of ABC Hardware Store and provided an automated approach to inventory and sales management.

5.2 CONCLUSIONS

A. What is a Conclusion?

- The Conclusion explains what can reasonably be concluded from the project's results.
- This is different from the Summary.

Summary: The system was developed and tested.

Conclusion:

- Based on the testing results, the developed system was able to automate the identified inventory processes and address the major problems encountered by the client.
- The conclusion should be based on evidence.

B. The Most Important Rule in Writing Conclusions

- DO NOT introduce new information.

Your conclusions must come from:

- objectives;
- findings;
- testing results;
- evaluation results; and
- observations documented in the project.

A useful formula is: **OBJECTIVE + RESULT + INTERPRETATION = CONCLUSION**

C. Conclusion Must Correspond to the Objectives

- This is one of the most important things students should understand.
- If Chapter 1 contains five objectives, Chapter 5 should normally provide conclusions that correspond to those objectives.

For Example:

Objective 1 :

“To automate inventory recording.”

Conclusion :

“The developed system successfully automated inventory recording by allowing authorized employees to add, update, search, and monitor product records electronically.”

Objective 2

“To automate sales transactions.”

Conclusion :

“The system successfully automated sales transactions by computing transaction totals and automatically updating inventory quantities after each completed sale.”

Objective 3

“To generate reports.”

Conclusion

“The developed system successfully generated inventory and sales reports, allowing authorized users to retrieve organized information for monitoring and decision-making.”

Sample Conclusion : “Using the ABC Hardware Store scenario”

Based on the results of the project, the researchers conclude that the developed Web-Based Inventory and Sales Management System successfully addressed the major problems identified in the organization's existing manual process. The system provided an automated mechanism for recording products, monitoring inventory levels, processing sales transactions, and generating reports.

The system was able to provide the required functionality identified during the requirements analysis. The implementation of user authentication and role-based access also provided controlled access to system functions according to the responsibilities of the users.

The results of functional testing demonstrated that the major modules performed according to their specified requirements. The successful execution of the test cases indicates that the system was capable of performing its intended operations under the conditions tested.

Furthermore, the user evaluation results indicated that the system was acceptable to its intended users based on the evaluation criteria used in the study. The findings therefore support the conclusion that the developed system provided a practical computing solution to the identified inventory and sales management problems of the organization.

Overall, the project objectives were substantially achieved through the successful development, implementation, testing, and evaluation of the proposed system.

5.3 RECOMMENDATIONS**A. What is a Recommendation?**

- Recommendations identify reasonable actions that can be taken after the completion of the project.

Recommendations may involve:

- system enhancements;
- additional features;
- future technologies;

- security improvements;
- additional integrations;
- broader implementation;
- additional testing;
- future research.

B. Recommendations Must Come From the Project

- Recommendations should not be random suggestions.

They should be based on:

- limitations encountered;
- user feedback;
- testing results;
- client requests;
- technologies not included in the current scope;
- future needs of the organization.

C. Example Scenario

Suppose the current inventory system does NOT have:

- mobile application;
- SMS notifications;
- barcode scanner;
- AI forecasting;
- cloud deployment;
- online payment.

Students may recommend these features if they are relevant to the organization's future needs.

Sample Recommendations :

1. Mobile Application

- It is recommended that a mobile version of the system be developed to allow authorized personnel to monitor inventory and sales transactions using smartphones and tablets.

2. Barcode Integration

- The system may be enhanced by integrating barcode scanning to improve the speed and accuracy of product identification during inventory and sales transactions.

3. Automated Notifications

- Future developers may implement email or SMS notifications to alert authorized personnel when product quantities reach their predefined reorder levels.

4. Cloud Deployment

- The system may be deployed in a cloud environment to provide secure remote access and centralized data management, subject to the organization's infrastructure and security requirements.

5. Business Analytics

- Future researchers may integrate data analytics and visualization tools to provide management with more comprehensive information for business decision-making.

E. Recommendations Should Be Specific

Weak

- Improve the system.

Better

- Future developers may enhance the system by adding a barcode scanning module that allows users to identify products automatically during sales transactions.

Weak

- Make the system more secure.

Better

- Future developers may strengthen security by implementing multi-factor authentication, password policies, audit logs, and periodic security assessment.

Weak

- Add AI.

Better

- Future researchers may integrate demand forecasting using historical sales data to predict future product requirements and support inventory planning.

F. Recommendations Should Be Realistic

Students should consider:

- budget;

- available technology;
- client capability;
- technical feasibility;
- project scope.

G. Recommendations for Different Stakeholders

- Students may organize recommendations according to the people who may benefit from them.

For the Client

- The organization is recommended to provide regular user training and establish a system maintenance schedule to ensure continued and effective use of the developed system.

For System Administrators

- System administrators should regularly back up the database and monitor user accounts to maintain data availability and security.

For Future Developers

Future developers may integrate barcode scanning and automated notifications to improve inventory monitoring.

For Future Researchers

- Future researchers may investigate the use of machine learning techniques for inventory demand forecasting.

Sample Recommendation :

Based on the findings and conclusions of the study, the researchers recommend the following:

1. Mobile Application Development. Future developers may develop a mobile version of the system to allow authorized users to monitor inventory and sales information through smartphones and tablets.

2. Barcode Integration. The system may be enhanced with barcode scanning functionality to improve product identification and reduce manual encoding during inventory and sales transactions.

3. Automated Notifications. Future versions may include email or SMS notifications for low-stock products to assist employees in monitoring inventory levels.

4. Enhanced Security. Future developers may implement additional security mechanisms such as multi-factor authentication, audit trails, and periodic vulnerability assessment.

5. Advanced Analytics. Future researchers may integrate data analytics and machine learning techniques to generate sales trends and inventory demand forecasts.

6. Continuous Maintenance and Training. The client is encouraged to conduct regular database backups, software maintenance, and user training to ensure the continued reliability and effective use of the system.